Task 2:

Code:

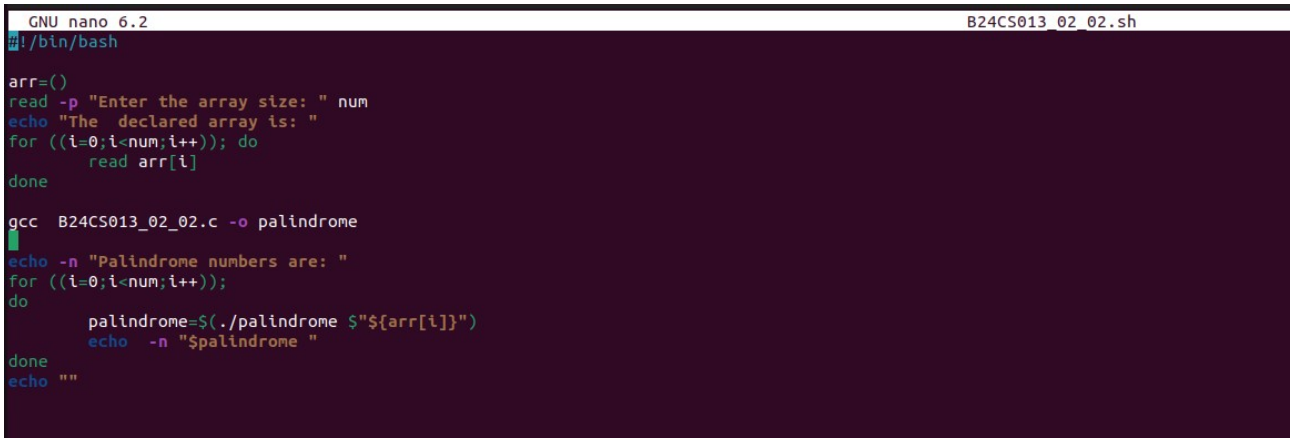
B24CS013_02_02.c



```
 GNU nano 6.2                                                    B24CS013_02_02.c
#include<stdio.h>
#include<stdlib.h>

int main(int argc,char *argv[]) {
if(argc<2) {
printf("Enter input!");
return 0;
}
int n=atoi(argv[1]);
if(n >=0 && n<=9) return n;
int m=n;
int rev=0;
while(m!=0) {
rev=rev*10+m%10;
m/=10;
}
if(rev==n) printf("%d",n);
else return 0;
}
```

B24CS013_02_02.sh



```
 GNU nano 6.2                                                    B24CS013_02_02.sh
#!/bin/bash

arr=()
read -p "Enter the array size: " num
echo "The  declared array is: "
for ((i=0;i<num;i++)); do
        read arr[i]
done

gcc  B24CS013_02_02.c -o palindrome

echo -n "Palindrome numbers are: "
for ((i=0;i<num;i++));
do
        palindrome=$(./palindrome $"${arr[i]}")
        echo  -n "$palindrome "
done
echo ""
```

Output:

```
user@user:~/Desktop/B24CS013_Bodike Chaithali_Exam_02$ nano B24CS013_02_02.sh
user@user:~/Desktop/B24CS013_Bodike Chaithali_Exam_02$ nano B24CS013_02_02.c
user@user:~/Desktop/B24CS013_Bodike Chaithali_Exam_02$ ls -l B24CS013_02_02.sh
-rwxrwxr-x 1 user user 333 Sep  9 16:41 B24CS013_02_02.sh
user@user:~/Desktop/B24CS013_Bodike Chaithali_Exam_02$ ls
B24CS013_02_02.c  B24CS013_02_02.sh
user@user:~/Desktop/B24CS013_Bodike Chaithali_Exam_02$ ./B24CS013_02_02.sh
Enter the array size: 5
The  declared array is:
12345
121
1287821
55464
223322
Palindrome numbers are:  121 1287821  223322
user@user:~/Desktop/B24CS013_Bodike Chaithali_Exam_02$ ls
B24CS013_02_02.c  B24CS013_02_02.sh  palindrome
user@user:~/Desktop/B24CS013_Bodike Chaithali_Exam_02$
```

Task 3:

```
user@user:~/Desktop/B24CS013_Bodike Chaithali_Exam_02$ sed 's/:/,/g' dir.txt
user1,CSE,student
user2,ME,faculty
user3,CSE,staff
user4,ME,student
user5,CSE,faculty
user@user:~/Desktop/B24CS013_Bodike Chaithali_Exam_02$
```

```
user@user:~/Desktop/B24CS013_Bodike Chaithali_Exam_02$ sed 's/:/,/g' dir.txt
user1,CSE,student
user2,ME,faculty
user3,CSE,staff
user4,ME,student
user5,CSE,faculty
user@user:~/Desktop/B24CS013_Bodike Chaithali_Exam_02$ cat dir.txt
user1:CSE:student
user2:ME:faculty
user3:CSE:staff
user4:ME:student
user5:CSE:faculty
user@user:~/Desktop/B24CS013_Bodike Chaithali_Exam_02$
```

Task 5:

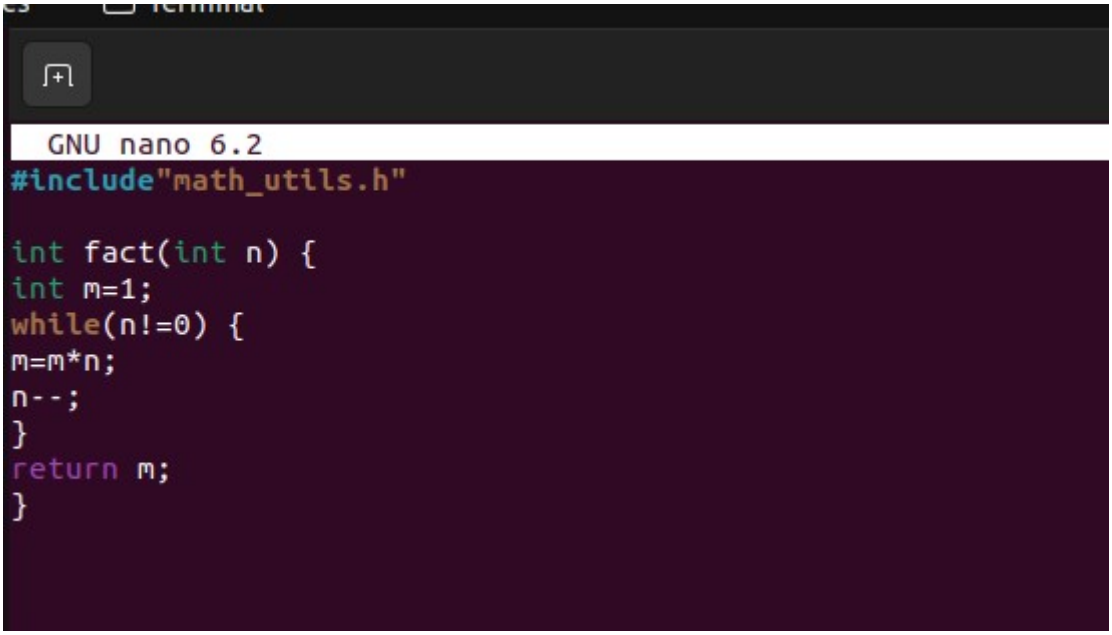
```
user@user:~/Desktop/B24CS013_Bodike Chaithali_Exam_02$ nano B24CS013_02_03.sh
user@user:~/Desktop/B24CS013_Bodike Chaithali_Exam_02$ mkdir B24CS013_LabExam02_05
user@user:~/Desktop/B24CS013_Bodike Chaithali_Exam_02$ ls B24CS013_LabExam02_05/
user@user:~/Desktop/B24CS013_Bodike Chaithali_Exam_02$ cd B24CS013_LabExam02_05/
user@user:~/Desktop/B24CS013_Bodike Chaithali_Exam_02/B24CS013_LabExam02_05$ ls
user@user:~/Desktop/B24CS013_Bodike Chaithali_Exam_02/B24CS013_LabExam02_05$ nano main.c
user@user:~/Desktop/B24CS013_Bodike Chaithali_Exam_02/B24CS013_LabExam02_05$ ls
main.c
user@user:~/Desktop/B24CS013_Bodike Chaithali_Exam_02/B24CS013_LabExam02_05$ nano math_utils.h
user@user:~/Desktop/B24CS013_Bodike Chaithali_Exam_02/B24CS013_LabExam02_05$ nano fact.c
user@user:~/Desktop/B24CS013_Bodike Chaithali_Exam_02/B24CS013_LabExam02_05$ nano fib.c
user@user:~/Desktop/B24CS013_Bodike Chaithali_Exam_02/B24CS013_LabExam02_05$ ls
fact.c  fib.c  main.c  math_utils.h
user@user:~/Desktop/B24CS013_Bodike Chaithali_Exam_02/B24CS013_LabExam02_05$ nano makefile
user@user:~/Desktop/B24CS013_Bodike Chaithali_Exam_02/B24CS013_LabExam02_05$ make
```

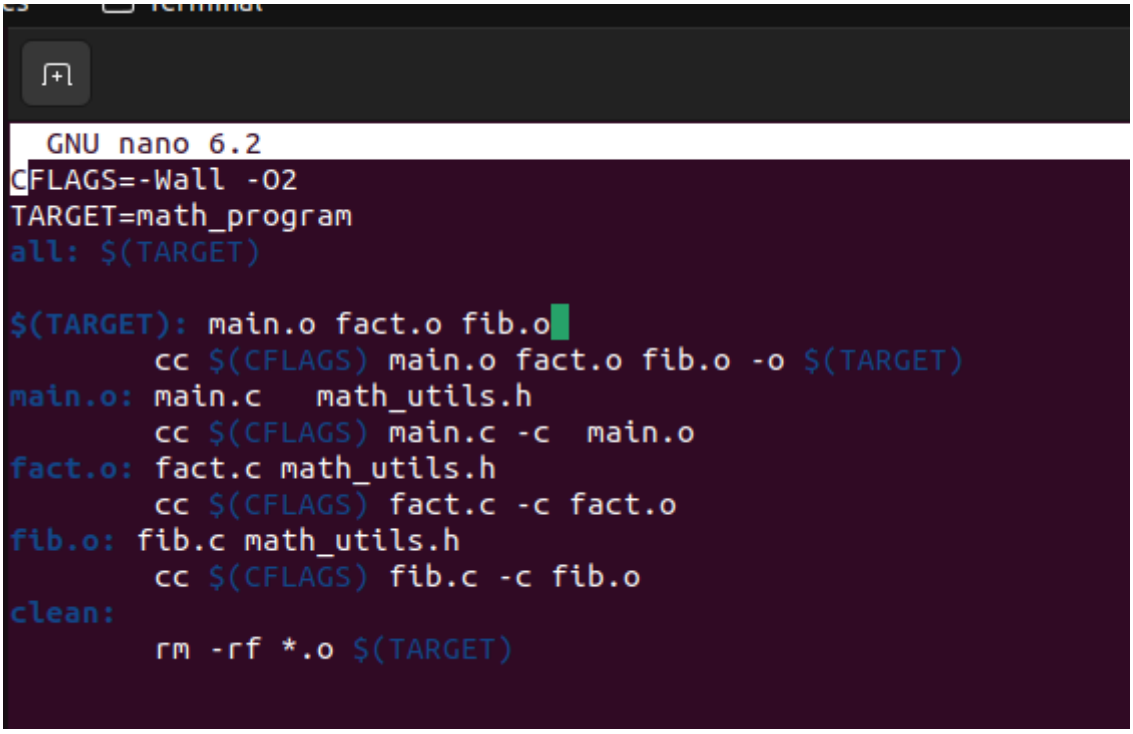
```
  GNU nano 6.2
#include"math_utils.h"

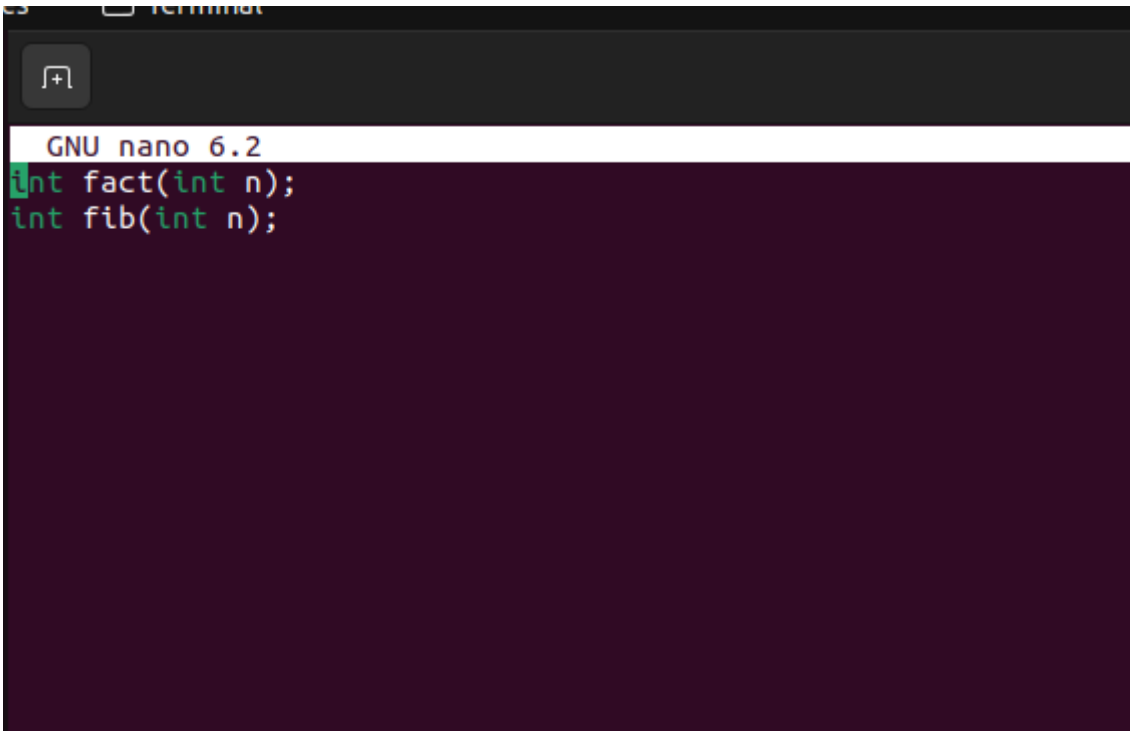
int fib(int n) {
int f1=0,f2=1,f3=f1+f2,i=0;
while(i<n) {
printf("%d ",f1);
f1=f2;
f2=f3;
f3=f1+f2;
i++;
}
return 0;
}
```

```
  GNU nano 6.2
#include<stdio.h>
#include"math_utils.h"

int main() {
int n;
printf("Enter any number: ");
scanf("%d",&n);
fib(n);
printf("\n%d!=%d\n",n,fact(n));
return 0;
}
```


GNU nano 6.2
#include"math_utils.h"
int fact(int n) {
int m=1;
while(n!=0) {
m=m*n;
n--;
}
return m;
}


GNU nano 6.2
CFLAGS=-Wall -O2
TARGET=math_program
all: $(TARGET)
$(TARGET): main.o fact.o fib.o
cc $(CFLAGS) main.o fact.o fib.o -o $(TARGET)
main.o: main.c math_utils.h
cc $(CFLAGS) main.c -c main.o
fact.o: fact.c math_utils.h
cc $(CFLAGS) fact.c -c fact.o
fib.o: fib.c math_utils.h
cc $(CFLAGS) fib.c -c fib.o
clean:
rm -rf *.o $(TARGET)



Q1)

```
user@user:~/Desktop/B24CS013_Bodike Chaithali_Exam_02/B24CS013_LabExam02_05$ make
cc -Wall -O2 main.c -c  main.o
main.c: In function ‘main’:
main.c:7:1: warning: ignoring return value of ‘scanf’ declared with attribute ‘warn_unused_result’ [-Wunused-result]
    7 | scanf("%d",&n);
      | ^~~~~~~~~~~~~~
cc: warning: main.o: linker input file unused because linking not done
cc -Wall -O2 main.o fact.o fib.o -o math_program
user@user:~/Desktop/B24CS013_Bodike Chaithali_Exam_02/B24CS013_LabExam02_05$ ./math_program
Enter any number: 5
0 1 1 2 3
5!=120
user@user:~/Desktop/B24CS013_Bodike Chaithali_Exam_02/B24CS013_LabExam02_05$ ls
fact.c  fact.o  fib.c  fib.o  main.c  main.o  makefile  math_program  math_utils.h
user@user:~/Desktop/B24CS013_Bodike Chaithali_Exam_02/B24CS013_LabExam02_05$ make clean
rm -rf *.o math_program
user@user:~/Desktop/B24CS013_Bodike Chaithali_Exam_02/B24CS013_LabExam02_05$ ls
fact.c  fib.c  main.c  makefile  math_utils.h
user@user:~/Desktop/B24CS013_Bodike Chaithali_Exam_02/B24CS013_LabExam02_05$
```